

Zhengyu (Byron) Zou

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EDUCATION

Brown University, Providence, RI — *Sc.B. in Mathematics, Sc.B. in Applied Mathematics–Computer Science*

GPA: 4.0/4.0 | *Expected Graduation: May 2026*

- **Advisors:** Richard Evan Schwartz, Bena Tshishiku, Caroline Klivans
- **Relevant Courses (“G” indicates graduate level classes):** Algebraic Geometry (G), Commutative Algebra (G), Algebraic Topology (G), Smooth Manifolds (G), Differential Geometry (G), Galois & Representation Theory, Matroid Theory, Real & Complex Analysis, Abstract Algebra, Partial & Ordinary Differential Equations, Graph Theory, Convex Analysis & Optimization (G), Stochastic Calculus, Statistical Inference (G), Data Structures & Algorithms, Theory of Computation, Algorithmic Machine Learning

RESEARCH EXPERIENCE

Brown University Department of Mathematics, Undergraduate Researcher

July 2025 – Present

- Advised by **Prof. Richard Evan Schwartz**
- Discovered a piecewise-isometric embedding of a 10-vertex (combinatorially minimal) geodesic triangulation of a hyperbolic 2-torus into hyperbolic 3-space with the aid of mathematical optimization.
- Very recently discovered a piecewise-isometric embedding with 12 vertices and 2-fold symmetry that implements the hyperelliptic involution. Investigating the points in the moduli space of 2-tori that are realized by this model.
- Paper preprint: <https://arxiv.org/abs/2509.18668>.
- Invited to seminar talks: UPenn Graduate Student Geometry and Topology Seminar (11/6/2025).
- Accompanying visualization program repository: <https://github.com/zzou9/hyperbolic-origami>.

Brown University Department of Mathematics, Undergraduate Researcher

June 2024 – December 2024

- Advised by **Prof. Richard Evan Schwartz**
- Prove the existence of invariant subsets of twisted polygons with precompact orbits under variants of the pentagram map, and discovered a birational formula that generalizes previous combinatorial formulas (projective geometry, geometric-inspired discrete integrable dynamical systems, combinatorics). Paper <https://arxiv.org/abs/2412.15561> to appear in *Arnold Mathematical Journal*.
- Developed an online simulation program using JavaScript to visualize pentagram map actions on projective polygons and compute relevant quantities. Access to program: <https://zzou9.github.io/pentagram-map>
- Invited to seminar talks: Hamiltonian Systems Seminar (3/18/2025), Young Mathematicians Conference (7/30/2025).

SEMINAR/CONFERENCES TALKS

University of Pennsylvania, Graduate Student Geometry and Topology Seminar

November 6th, 2025

- Seminar talk on the research paper “Vertex-Minimal Hyperbolic Origami 2-Torus.”
- Website: <https://www.math.upenn.edu/events/vertex-minimal-hyperbolic-origami-2-torus>.

Brown University, Undergraduate Math Colloquium

October 15th, 2025

- Seminar talk on the research paper “Vertex-Minimal Hyperbolic Origami 2-Torus.”
- Website: <https://www.math.brown.edu/clarso19/undergraduate-colloquium.html>.

Ohio State University, Young Mathematicians Conference

July 30th, 2025

- Seminar talk on the research paper “Spirals and the Dynamics of Deep Diagonal Maps.”
- Program website: <https://ymc.osu.edu/ymc-program-2025>.

University of Toronto, Hamiltonian Systems Seminar

March 13th, 2025

- Seminar talk on the research paper “Spirals, Tic-Tac-Toe Partition, and Deep Diagonal Maps.”
- Website: https://seminars.math.toronto.edu/pages/seminars?case=view_talk&talk_id=1742315400-1742297400-460.

Brown University, Undergraduate Math Colloquium

October 16th, 2024

- Seminar talk on the research paper “Spirals, Tic-Tac-Toe Partition, and Deep Diagonal Maps.”
- Website: <https://www.math.brown.edu/clarso19/undergraduate-colloquium.html>.

TEACHING EXPERIENCE

Brown University, Undergraduate Teaching Assistant

September 2023 – Present

- Undergraduate teaching assistant for Topology (MATH1710, fall 2025) under Prof. Willie Rush Lim
- Undergraduate teaching assistant for Real Analysis (MATH1630, fall 2024) under Prof. Justin Holmer
- Undergraduate teaching assistant for Algorithmic Machine Learning (CSCI1952Q, spring 2024) under Prof. Yu Cheng
- Undergraduate teaching assistant for Honors Multivariable Calculus (MATH0350, fall 2023) under Prof. Christine Breiner

Brown University Math Resource Center, Tutor

October 2022 – May 2023

- Organized weekly tutoring sessions to help students with problems in intro-level math classes (and sometimes more advanced topics)

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Self-Employed, *mathematics tutor*

July 2024 – Present

- Tutored more than 50 private lessons for more than seven students whose ages range from middle school to high school juniors. The topics include SAT preparation, algebra, geometry, math-modeling, calculus.

INTERNSHIP/VOLUNTEERING EXPERIENCE

Brown University Math Olympiad, *Problem Writer, Test Day Volunteer*

March 2025 – Present

- Wrote a total of 4 exam problems (one in use) for BrUMO competition. Attended weekly meetings to discuss/test-solve potential exam problems.

Walnut Education, *Teaching Assistant; Tutor; Software Engineer*

July 2023 – August 2023

- Assisted SAT and TOEFL students by organizing practice tests and vocabulary lessons; provided problem-solving support
- Developed a course scheduling and database management system using Flask and MySQL to manage information for over 50 students and employees

SKILLS & INTERESTS

Competitions: Putnam 2024 – top 300 (score: 29)

Programming: [*Advanced / Proficient*]: Matlab, Mathematica, Python, Java, C++, JavaScript, HTML, CSS, SQL, LaTeX .

Language: English (Native), Chinese (Native), Spanish (Proficient)

Interests: jazz drumming, long-distance running, traveling, food